

Summary

Design envelope (worst case, cornering):

$R_{upper} = 1113 \text{ N}$

$R_{lower} = 3069 \text{ N}$

Bump = 2207 N

Material: Al 6061-T6 ($S_y = 276 \text{ MPa}$)

Max von Mises stress: reduced from 22.7 MPa to 35MPa

Max deflection: reduced from 0.007 mm to 0.005mm

FOS = 8 compared to required 1.5, can be massively reduced in future design changes

Mass reduced 10–33% through topology optimization

Manufacturing: 3-axis CNC

Component fitting and assumptions

GIVEN (project spec)

Vehicle mass 300 kg, 50% front

Design accels: 2g braking, 1.5g lateral, 3g bump

Min FOS 1.5

SKF 6008-2Z bearing, Brembo P30, 16" Hoosier

MEASURED (reference geometry file)

Wheelbase 1593 mm Track width 1166 mm Wheel radius 203.2 mm

Hub =upper BJ 95 mm

Hub =lower BJ 95 mm

Kingpin axis: ΔX 7.53, ΔY 12.28, ΔZ 189.45 mm

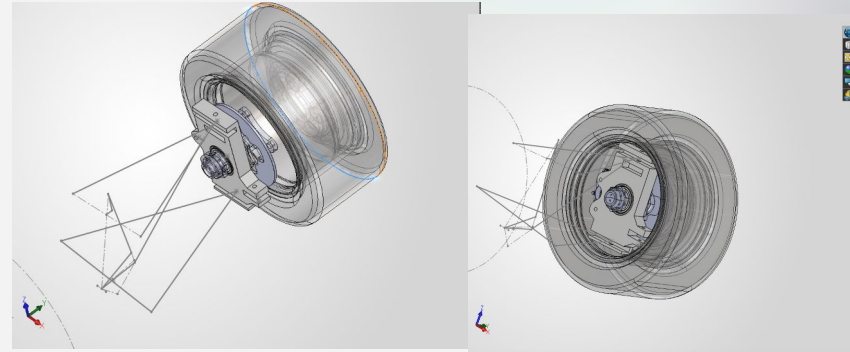
DERIVED

$$\text{KPI} = \arctan(\Delta Y / \Delta Z) = 3.71^\circ$$

$$\text{Caster} = \arctan(\Delta X / \Delta Z) = 2.28^\circ$$

ASSUMED

CG height 300 mm =not specified by the team.

[illegible]

Calculations details

1. Steril



assume parallel



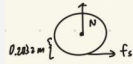
$$N \Rightarrow \frac{1}{4}(300 \times g) = \frac{300g}{4} = 75g \text{ N}$$

2. Braking (pitch)

$$\Delta F = \frac{M}{L} = \frac{m a h_{cg}}{L}$$

$$= \frac{(300)(29)(0.300)}{(1.59346)} = 1108 \text{ N for both tire-tires}$$

$$\Delta F = 554 \text{ N per tire}$$



$$N_{\text{brake}} = 75g + \frac{1104}{2} \approx 1289.8 \text{ N}$$

$$F = (m a_{max})$$

$$(m a_{max}) = \frac{m v}{t_{brake}} = f_k$$

$$\frac{m v}{t_{brake}} = \mu N = \mu m g$$

$$\frac{v}{t_{brake}} = \mu g$$

$$t_{brake} = \frac{v}{\mu g}$$

$$t_{brake} = \frac{20}{0.8 \times 9.8}$$

$$t_{brake} = 2.55 \text{ s}$$

$$f_{kr} \cdot N_{brake} = 2 \times 1289.9 \approx 2579.7 \text{ N}$$

Brake torque (caliper)

$$M_{fk}^{(c)} = M - f_k$$

$$= (0.1032 \text{ cy}) \times (3579.7 \text{ N/cy})$$

$$= -514.2 \text{ N}$$

4. Bumping



$m_1 = 3g(300) = 900g$ whole car

$$\frac{9009}{4} \approx 2207.25 \text{ N}$$

3 cornering case (roll)

t = track width

$$\Delta N = \frac{mg(1 + \frac{v^2}{c^2})}{1} = \frac{(800 \text{ kg} \times 9.8) (1 + \frac{(2300)^2}{c^2})}{1.16628} \approx 1125.53$$

$$N_{\text{curier}} = 25g + 1135.5 \approx 1160.5g$$

$$\mu = \frac{1.5g}{g} = 1.5$$

$$f_k = 1.5 (1871.28) \approx 2806.92 \text{ N}$$

Moment

$$r \times (F_1) + -r \times (F_2) - A \times (f_K) = 0$$

$$r(F_1) + (F_1 + f_k)r = R(f_k)$$

$$2F_r + f_{kr} = R(f_k)$$

$$F_1 = \frac{f_R \cdot (R-r)}{2r}$$

$$= \frac{(1106)(0.1032 - 0.095)}{2(0.095)} = 1597.94 \text{ N}$$

$$F_1 + F_2 = -f_k$$

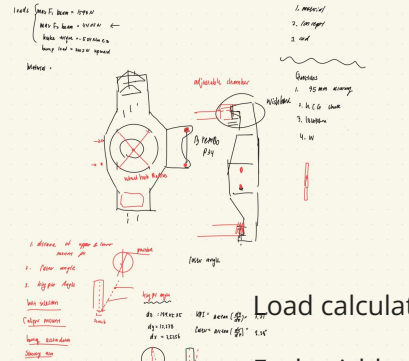
$$F_2 = -2806 - (1597.94)$$

$$= -4405.9 \text{ N}$$

$$F_{\text{net}} \Rightarrow F_1 + F_2 = -f_k$$

$$F_1 + f_K = -F_2$$

design load (inside wheel life)



Load calculation method

Each wishbone leg and the tie rod is a two-force member
load is purely axial along the joint-to-joint line.

Upright modeled as a beam along the kingpin axis:

$$R_{upper} = F \cdot (r_{wheel} - a_l) / (a_u + a_l)$$

$$R_{\text{lower}} = -F \cdot (r_{\text{wheel}} + a_u) / (a_u + a_l)$$

$a_u = 95 \text{ mm}$, $a_l = 95 \text{ mm}$, $r_{\text{wheel}} = 203.2 \text{ mm}$

Effective friction coefficient taken directly from the specified accelerations: $\mu = a/g$

Braking and cornering treated as separate governing cases, the tire friction circle physically prevents both at full magnitude.

Calculations ([link](#))

- 5 different situations

Google
Sheet
validation

case 1: static			
normal force(N)	735.75		
case 2: braking	two wheel	individual wheel	friction_coefficient
load transfer(N)	1108.154582	554.0772909	2
Normal	1,289.83		
Frictional force	2579.654582		
Brake torque	-524.185811		
case 3: cornering (roll)		friction_coefficient	individual car
change in Normal	1135.533491	1.5	567.7667455
Normal	1,303.52		
frictional force	1955.275118		
A arm upper	1113.477725		
A arm lower	-3068.752844		
case4: bumping case			
a= 3g			
vertical force	2207.25		

Manufacturing

- Axis cnc 3 axis
- All internal corner and pockets to a standard end mill diameter of R6
- No features needed to get from the sides
- Went from a curved top view surface to straightline for cheaper and more simplistic option

A-arm, tie rod, material and screw selection

Material, bearing and hardware selection

MATERIAL

- Al 6061-T6 (baseline)

Sy 276 MPa, Su 310 MPa, E 68.9 GPa, ρ 2700 kg/m³

Weldable, lowest cost, most common FSAE upright material.

BEARING

- SKF 6008-2Z

40 mm bore, 68 mm OD, 15 mm width. Static rating 11 kN.

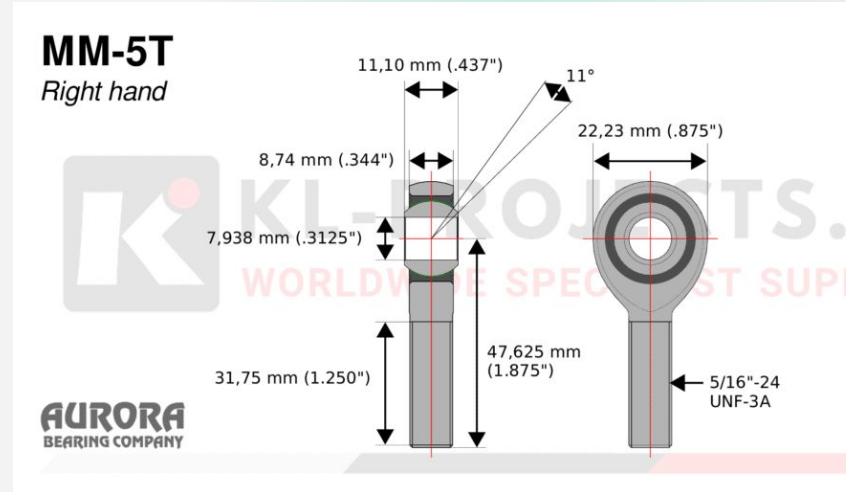
Bore in the upright is designed around the 68 mm OD seat.

WISHBONE HARDWARE

- Aurora MM-T5 rod end

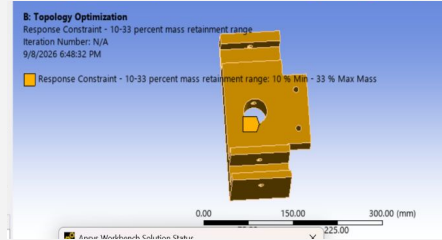
FASTENERS

- per FSAE T.8: SAE Grade 5 / Metric Class 8.8 minimum, positive locking, min two full threads projecting past each lock nut. Rod ends in double shear per V.3.1.5

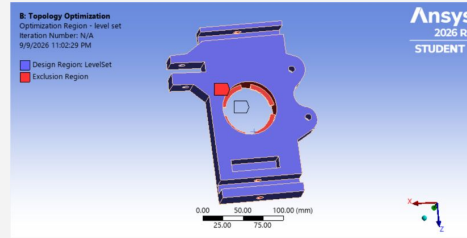


Topology optimizations testing/considerations

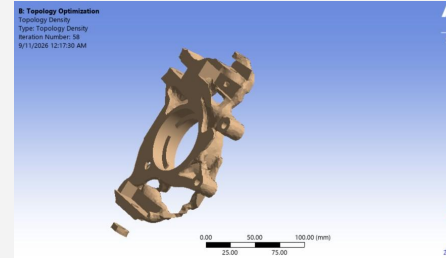
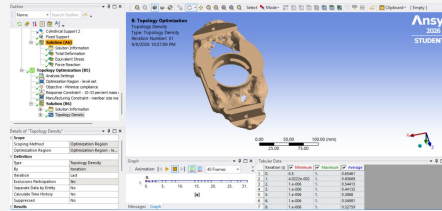
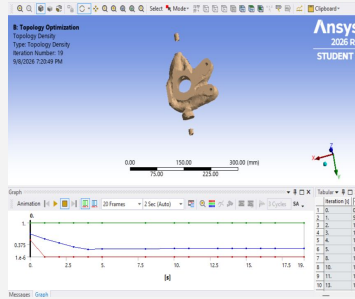
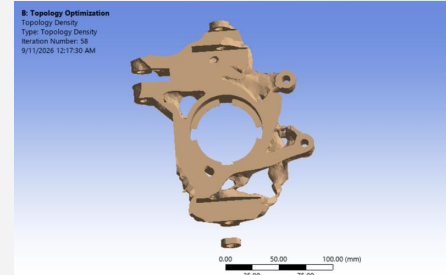
100 interactions, 4mm mesh, 2mm exclusion



200 interactions, 3mm mesh, 5mm exclusion
- tie rod and caliper was erroneously excluded



300 interactions, 3mm mesh, 5mm exclusion
10-33% reduction in mass



1 Block-out

Bounding volume from hardpoints and keep-out regions.

All mounting features located, no material removed.

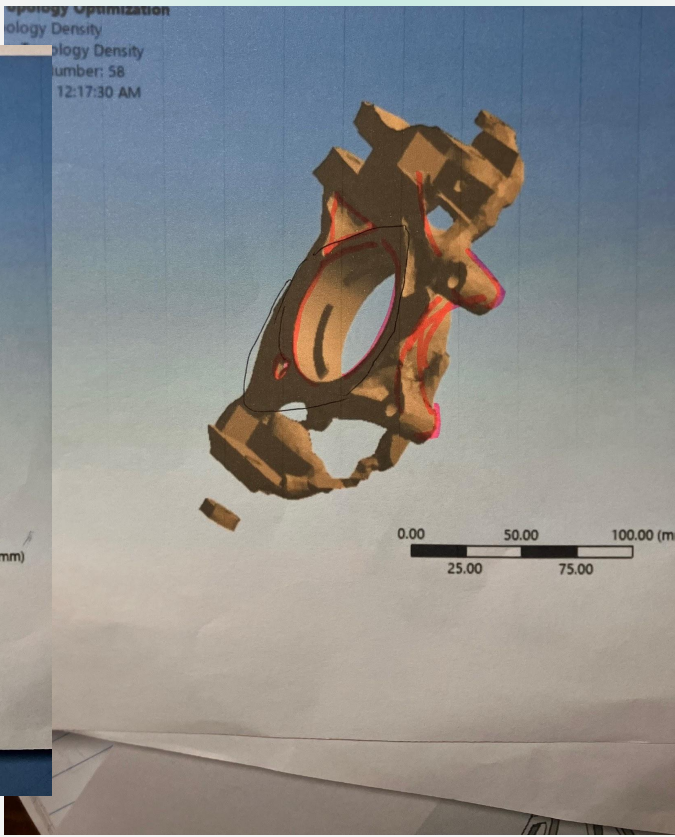
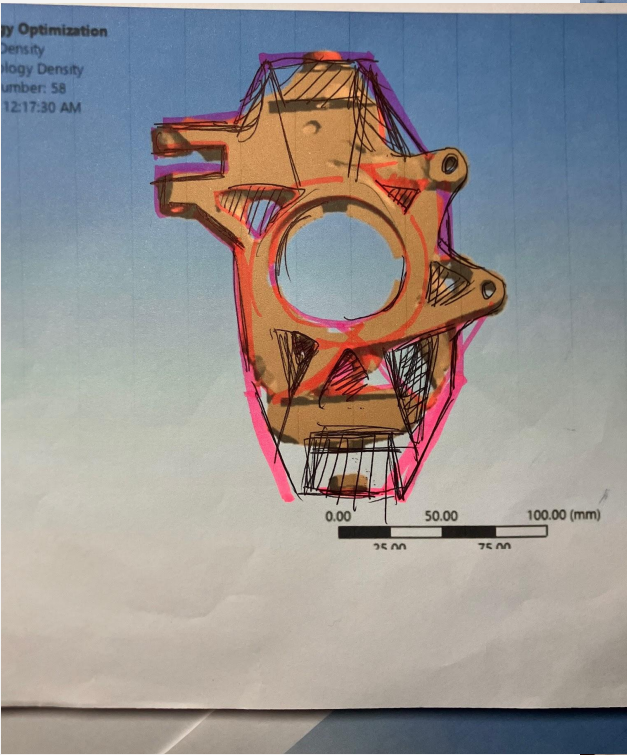
2 First basic test topology run

Result non manufacturable organic geometry, floating island

3 Re-interpretation and drafting reconsidered in deisgn

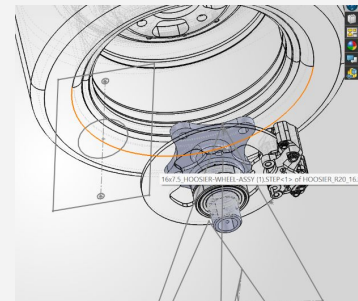
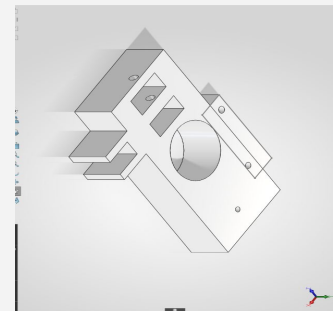
4 Final

Triangulated web between upper boss, bearing bore and lower boss. Pockets sized to a standard end mill.

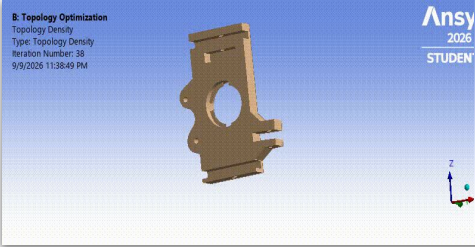
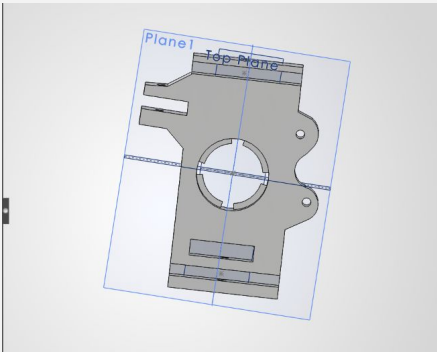


Design iterations

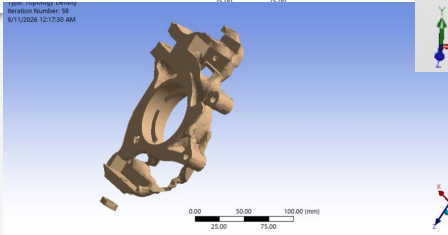
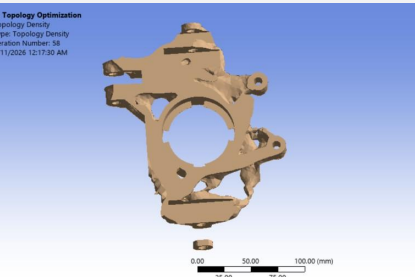
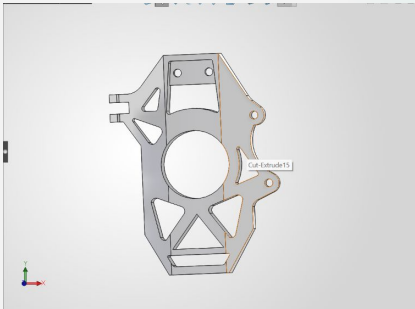
Basic run



Second



Third



Final

